Agents in AI

How software learned to act, not just answer

What Makes Software an Agent

Perceive

Read its environment.

Reason

Plan a course of action.

Act

Use tools to change things.

Assistant vs Agent

Assistant	Agent
Answers one prompt	Pursues a goal
Waits for you	Acts on its own
Has no tools	Calls tools

Core Capabilities

Capability	What it does
Planning	Break goals into steps
Tool Use	Call APIs and code
Memory	Recall past context
Reflection	Critique and retry

Why Now

2022

LLMs Arrive

tool use emerges

10x

Faster

early reports

1M+

Developers

building agents

From Rules to Agents

1980s	2010s	Today
Rules	**ML**	**Agents**
Hand-coded.	Learned patterns.	Models that act.

Common Agent Patterns

Pattern	In one line
ReAct	Reason, then act
Tool Use	Call functions
RAG	Retrieve facts first
Multi-Agent	Specialists team up

Multi-Agent Systems

Many narrow agents beat one generalist.

Each owns a clear role.

A coordinator routes the work.

They hand off context.

Promise vs Pitfalls

Promise

Automate real work

Scale expertise

Run all day

Pitfalls

Errors compound

Hard to predict

Needs guardrails

Takeaways

Software that acts, with human oversight